OFFICIAL COURSE PROBLEM SOLUTIONS • ZERO SKIPPED STEPS

Topic 3: Financial Statements (Exercises 1 to 6)

Source Document: `T3_Financial Statements_EXERCISES.pdf` (UC3M Business Management) • Solved with exhaustive mathematical and accounting steps, exam-day strategies, and balanced statements.

Method used in every solution

Central idea: every operation of an exam problem is read **three times**: does it create a **revenue or expense** of the period (income statement)? does **cash** enter or leave before the closing date (cash account)? what **asset, liability or equity** item remains open at the closing date (balance sheet)? If the three readings are done for every operation, the balance sheet closes by itself.

The full step-by-step method (classification table, depreciation and interest rules, cash account and balance-sheet layout) is at the top of the exam-type problems page. Each solution below follows the same four phases: setup, frames (period and classification of every operation), derivation, interpretation.

Exercise 1

Exercise 1 — Revenues vs collections, expenses vs payments, and depreciation

Original statement · T3_Financial Statements_EXERCISES.pdf

A company started activity on 1 January of the present year. At 31 March it has the following goods, acquired on 31 December of the previous year:

- *factory building 1,000 m.u., duration 25 years;*
- *office building 1,500 m.u., duration 50 years;*
- *machinery 300 m.u., duration 10 years;*
- *transport equipment 100 m.u., duration 5 years.*

Information for the first quarter:

- *personnel salaries of the quarter 450 m.u., of which 75 m.u. are pending to be paid after 31/3;*
- *materials purchased and used in production 700 m.u., of which 125 m.u. are pending to be paid after 31/3;*
- *energy consumption estimated at 230 m.u., nothing paid to date;*

- *sales invoiced to customers 1,930 m.u., of which 315 m.u. had not been collected at 31/3; the rest was collected in due time. In manufacturing all the units sold, all the purchased materials were consumed.*

Ignore corporation taxes.

a) At 31/3, find the income, expenses, collections (cash inflows), payments (cash outflows), investments and depreciation of the quarter (linear depreciation along the lifetime of the goods).

b) How does the answer change if only half of the purchased materials had been consumed in manufacturing the units sold?

Phase 1 — Setup: data and hypotheses

Period: first quarter (1/01 to 31/03), $\frac{3}{12} = \frac{1}{4}$ of a year. The fixed assets were bought on 31/12 of the previous year, so they are used during the **whole** quarter.

The exercise trains the distinction of [Slide 8]: a **revenue** is recorded when the sale is made, whether or not it is collected; an **expense** when the good or service is received or consumed, whether or not it is paid. Collections and payments are the cash side of the same operations.

Phase 2 — Frames: classification of every operation

Operation	Revenue / expense of Q1	Collection / payment in Q1	Pending at 31/3
Sales	Revenue 1,930	Collection $1{,}930 - 315 = 1{,}615$	Receivable 315
Salaries	Expense 450	Payment $450 - 75 = 375$	Salaries payable 75
Materials (all consumed)	Expense 700	Payment $700 - 125 = 575$	Supplier payable 125
Energy	Expense 230	Payment 0	Energy payable 230
Fixed assets	Depreciation 30	0	Net value $2{,}900 - 30$

Phase 3 — Derivation

Step 1. Depreciation of the quarter

For each asset, with no residual value given:

$$D_{\text{quarter}} = \frac{\text{Cost}}{\text{Years of life}} \times \frac{3}{12}$$

Asset	Annual depreciation	Quarter
Factory building	$1{,}000/25 = 40$	$40/4 = 10.0$
Office building	$1{,}500/50 = 30$	$30/4 = 7.5$
Machinery	$300/10 = 30$	$30/4 = 7.5$
Transport equipment	$100/5 = 20$	$20/4 = 5.0$
Total	**120**	**30.0**

Step 2. Part (a)

- **Income (revenues):** the sales invoiced, **1,930 m.u.**
- **Expenses:** $450 + 700 + 230 + 30 =$ **1,410 m.u.** (so the profit of the quarter is $1{,}930 - 1{,}410 = 520$ m.u.).
- **Collections:** $1{,}930 - 315 =$ **1,615 m.u.**
- **Payments:** $(450 - 75) + (700 - 125) + 0 = 375 + 575 =$ **950 m.u.** Depreciation is not a payment.
- **Investments of the quarter: 0 m.u.** The fixed assets (total $1{,}000 + 1{,}500 + 300 + 100 = 2{,}900$ m.u.) were acquired on 31/12 of the previous year, before the quarter began. If the question is read as "the investment the company works with", the answer is that 2,900 m.u.; state the reading you use in the exam.
- **Depreciation: 30 m.u.**

Step 3. Part (b): only half of the materials consumed

Only the material **consumed** is an expense [Slide 8, Slide 10]: $700/2 = 350$ m.u. The other 350 m.u. stay in the warehouse as **inventory**, a current asset [Slide 14].

Magnitude	(a)	(b)	Change
Income	1,930	1,930	0
Expenses	1,410	$450 + 350 + 230 + 30 = 1{,}060$	−350
Profit	520	870	+350
Collections	1,615	1,615	0
Payments	950	950	0
Inventory at 31/3	0	350	+350
Depreciation	30	30	0

Payments do not change: the company **paid** for what it **bought** (575 of the 700), regardless of how much it **used**.

Phase 4 — Interpretation and checks

- **Profit is not cash:** in (a) the profit is 520 while the net cash generated is $1{,}615 - 950 = 665$. The difference, $665 - 520 = 145$, is explained item by item: $+30$ (depreciation, an expense without payment) $+75 + 125 + 230$ (expenses not yet paid) -315 (revenue not yet collected) $= 145$.
- **Part (b)** raises the profit by 350 without any extra cash: unconsumed material is an asset, not an expense.
- **Check of depreciation:** the annual total 120 times $\frac{1}{4}$ gives 30.

Check against the official key

OFFICIAL SOLUTION, T3_FINANCIAL STATEMENTS_SOLUTIONS.PDF, PP. 1-2

Part (a) coincides with the key: income 1,930; collections 1,615; payments 950; expenses 1,410; depreciation 30; investments 0. In part (b) the key also states that revenue, collections and payments do not change and that the unused material appears as inventory.

ERRATUM IN THE OFFICIAL SOLUTION (PART B)

The key gives total expenses of 1,124 for part (b), but its own items add to $450 + 350 + 230 + 30 = 1{,}060$. That is the value obtained above, and the only one consistent with a profit that rises by exactly the 350 of material that was not consumed.

Exercise 2

Exercise 2 — JUGA S.A., opening balance sheet

Original statement · T3_Financial Statements_EXERCISES.pdf

> *The JUGA S.A. firm is started with a shareholders' contribution of 60,000 €. Of those, 30,000 € are paid in and deposited in a bank account, and the remaining 30,000 € correspond to the transfer of a patent. With the cash of the bank account, raw materials are purchased for 10,500 € and machinery for 15,000 €. Detail the balance sheet at the start of operations.*

Phase 1 — Setup: data and hypotheses

- The capital contributed is **60,000 €**: 30,000 € in money and 30,000 € in kind (a patent).
- The purchases are paid **with the bank account** (no debt is created).
- It is the **start** of operations: nothing has been produced or sold yet, so there are no revenues or expenses and no income statement.

Phase 2 — Frames: effect of each operation on the balance sheet

Operation	Assets	Liabilities	Equity
Money contributed	Bank +30,000		Capital +30,000
Patent contributed	Patent +30,000		Capital +30,000
Raw materials bought	Raw materials +10,500; bank −10,500		
Machinery bought	Machinery +15,000; bank −15,000		

Each row keeps Assets = Liabilities + Equity [Slide 12]: the contributions increase both sides; the purchases only change the composition of the assets.

Phase 3 — Derivation

Step 1. Bank balance

$$\text{Bank} = 30{,}000 - 10{,}500 - 15{,}000 = 4{,}500\ €$$

Step 2. Classification

- **Patent:** non-current asset [Slide 15].
- **Machinery:** non-current asset (equipment) [Slide 15].
- **Raw materials:** inventory, a current asset [Slide 14]; not consumed yet, so not an expense.
- **Bank:** cash, a current asset [Slide 14].
- **Share capital:** contributed capital [Slide 17].

Step 3. Balance sheet at the start of operations

Assets	**€**	**Liabilities and equity**	**€**
Patent	30,000	Share capital	60,000
Machinery	15,000	**Equity**	**60,000**
Non-current assets	**45,000**	**Liabilities**	**0**
Raw materials	10,500		
Bank	4,500		
Current assets	**15,000**		
Total assets	**60,000**	**Total liabilities and equity**	**60,000**

Phase 4 — Interpretation and checks

- **Balance:** $45{,}000 + 15{,}000 = 60{,}000 = 60{,}000 + 0$.
- **Typical mistake:** keeping the bank at 30,000 € while also adding the machinery and the raw materials gives assets of 85,000 € against 60,000 € of equity. Spending cash on an asset never changes the total.
- **Financial structure** [Slide 36]: the company is financed 100% with equity ($L/E = 0$).

Check against the official key

OFFICIAL SOLUTION, T3_FINANCIAL STATEMENTS_SOLUTIONS.PDF, P. 3

The balance sheet coincides with the key in every line: bank 4,500; patent 30,000; machinery 15,000; raw materials 10,500; share capital 60,000; total 60,000. The bank movements (+30,000, −15,000, −10,500) are the same.

Exercise 3

Exercise 3 — Libby's Company, balance sheet from a list of accounts

Original statement · T3_Financial Statements_EXERCISES.pdf, p. 2

Prepare the balance sheet of Libby's Company using the following figures (in euros):

Account	*€*	*Account*	*€*
Short-term (current) loan debts	*15,500*	*Accounts payable*	*1,500*
Long-term loan debts	*3,000*	*Equipment*	*13,000*
Money on its bank account	*3,000*	*Raw materials (purchased this year but not used yet)*	*12,500*
Cash	*500*	*Furniture*	*1,500*
Paid-in capital	*12,000*	*Retained earnings from the year*	*2,000*
Accounts receivable	*3,500*		

Phase 1 — Setup: what has to be done

Every account is already a balance-sheet item: there are no revenues or expenses. The task is pure **classification** into the five blocks of [Slide 18] and a check that the totals agree.

Phase 2 — Frames: classification of every account

Account	Block	Reason
Equipment	Non-current asset	Long-lived resource [Slide 15]
Furniture	Non-current asset	Long-lived resource [Slide 15]
Raw materials (not used yet)	Current asset (inventory)	Will be used within the year [Slide 14]
Accounts receivable	Current asset	Turned into cash within the year [Slide 14]
Bank account	Current asset (cash)	[Slide 14]
Cash	Current asset (cash)	[Slide 14]
Paid-in capital	Equity: contributed capital	[Slide 17]
Retained earnings from the year	Equity: retained earnings	Earnings not paid as dividends [Slide 13]
Long-term loan debts	Long-term liability	Not paid within the year [Slide 16]
Short-term loan debts	Current liability	Paid within the year [Slide 16]

Account	Block	Reason
Accounts payable	Current liability	Owed to suppliers, short term [Slide 13, Slide 16]

Phase 3 — Derivation

Step 1. Block totals

$$\text{Non-current assets} = 13{,}000 + 1{,}500 = 14{,}500$$

$$\text{Current assets} = 12{,}500 + 3{,}500 + 3{,}000 + 500 = 19{,}500$$

$$\text{Equity} = 12{,}000 + 2{,}000 = 14{,}000$$

$$\text{Long-term liabilities} = 3{,}000, \qquad \text{Current liabilities} = 15{,}500 + 1{,}500 = 17{,}000$$

Step 2. Balance sheet

Assets	€	Liabilities and equity	€
Equipment	13,000	Paid-in capital	12,000
Furniture	1,500	Retained earnings	2,000
Non-current assets	**14,500**	**Equity**	**14,000**
Raw materials	12,500	Long-term loan debts	3,000
Accounts receivable	3,500	**Long-term liabilities**	**3,000**
Bank account	3,000	Short-term loan debts	15,500
Cash	500	Accounts payable	1,500
Current assets	**19,500**	**Current liabilities**	**17,000**
Total assets	**34,000**	**Total liabilities and equity**	**34,000**

Phase 4 — Interpretation and checks

- **Balance:** $14{,}500 + 19{,}500 = 34{,}000 = 14{,}000 + 3{,}000 + 17{,}000$.
- **Financial structure** [Slide 36]: liabilities $L = 3{,}000 + 17{,}000 = 20{,}000$, equity $E = 14{,}000$, $E + L = 34{,}000$:

$$\%E = \frac{14{,}000}{34{,}000} = 41.2\%, \qquad \%L = \frac{20{,}000}{34{,}000} = 58.8\%, \qquad \frac{L}{E} = \frac{20{,}000}{14{,}000} = 1.43$$

The company has more debt than equity, and most of it (17,000 of 20,000) is due within the year: by [Slide 36], a high leverage indicates higher risk to shareholders.

- **Typical mistakes:** retained earnings are equity, not a liability; accounts receivable (owed **to** the company) are an asset, accounts payable (owed **by** the company) a liability.

Check against the official key

OFFICIAL SOLUTION, T3_FINANCIAL STATEMENTS_SOLUTIONS.PDF, P. 4

The balance sheet coincides with the key in every block: current assets 19,500; non-current assets 14,500; current liabilities 17,000; non-current liabilities 3,000; owners' equity 14,000; total 34,000. The key asks only for the balance sheet, so the leverage figures in Phase 4 are an addition based on Slide 36.

Exercise 4

Exercise 4 — Hawkins Partners, income statement with inventories

Original statement · T3_Financial Statements_EXERCISES.pdf, p. 2

Prepare the income statement for Hawkins Partners using the following figures (in euros), knowing that the applicable corporate tax rate is 30%:

#	Item	€	#	Item	€
1	*Initial inventory*	600	6	*Interests from the long-term debt*	5,500
2	*Purchase of goods (sold 100%)*	84,000	7	*Wages and salaries*	12,700
3	*Rental costs*	1,000	8	*Water, gas and electricity*	1,200
4	*Property sold to another company (income)*	10,500	9	*Sales*	130,800
5	*Interests received by the company*	15,200	10	*Inventory at the end of the period*	500

Phase 1 — Setup: what each figure is

The statement mixes the three kinds of income-statement lines of [Slide 9]: items of the primary operation (sales, goods, wages, rent, utilities), **non-operating items** (interests, the sale of property) [Slide 11], and the tax. It also gives the initial and final **inventory**, which adjust the cost of goods sold.

Phase 2 — Frames: classification of every item

Item	Line of the income statement
9. Sales	Sales revenue
1, 2, 10. Inventories and purchases	Cost of goods sold
3, 7, 8. Rent, wages, utilities	Operating expenses [Slide 10]
4. Property sold (income)	Non-operating item: gain on the sale of property [Slide 11]
5. Interests received	Non-operating item: interest revenue [Slide 8, Slide 11]
6. Interests on long-term debt	Non-operating item: interest expense [Slide 11]
Tax 30%	Income taxes [Slide 11]

Phase 3 — Derivation

Step 1. Cost of goods sold

The goods sold are those available (initial inventory + purchases) minus those still in stock at the end:

$$\text{COGS} = 600 + 84{,}000 - 500 = 84{,}100\,€$$

The inventory fell by $600 - 500 = 100$ €: those 100 € of goods were also sold, so they add to the purchases.

Step 2. Income statement

Line	Computation	€
Sales revenue		130,800
– Cost of goods sold	Step 1	84,100
= Gross profit	$130{,}800 - 84{,}100$	**46,700**
– Wages and salaries		12,700
– Rental costs		1,000
– Water, gas and electricity		1,200
= Income from operations	$46{,}700 - 14{,}900$	**31,800**
+ Gain on property sold		10,500
+ Interests received		15,200
– Interests on long-term debt		5,500
= Income before taxes	$31{,}800 + 10{,}500 + 15{,}200 - 5{,}500$	**52,000**
– Income taxes (30%)	$0.30 \times 52{,}000$	15,600
= Net income	$52{,}000 - 15{,}600$	**36,400**

Phase 4 — Interpretation and checks

- **Non-operating items add 20,200 €** ($10{,}500 + 15{,}200 - 5{,}500$): almost 39% of the income before taxes does not come from the primary operation, which is why the slides separate them [Slide 11].
- **The tax base** is the income before taxes (52,000), not the income from operations [Slide 11].
- **Reading of item 4:** the statement labels the 10,500 € as income from the property sold; no book value of the property is given, so the whole amount is taken as the gain on the sale.
- **Check:** $130{,}800 + 10{,}500 + 15{,}200 = 156{,}500$ of revenues and gains; $84{,}100 + 14{,}900 + 5{,}500 = 104{,}500$ of expenses; $156{,}500 - 104{,}500 = 52{,}000$.

Check against the official key

OFFICIAL SOLUTION, T3_FINANCIAL STATEMENTS_SOLUTIONS.PDF, P. 5

Cost of goods sold 84,100 (purchases 84,000 plus the inventory variation of 100), operating expenses 14,900, interests received 15,200, interests paid 5,500, income before taxes 52,000, tax 15,600 and net income 36,400 coincide with the key.

DISCREPANCY WITH THE OFFICIAL SOLUTION (EBIT)

The key includes the 10,500 € of the property sold (labelled "Income from rentals") in the income from operations, so its EBIT is 42,300 €. Here the amount is shown below EBIT, as a non-operating gain, following the definition of Slide 11 (gains and losses on the sale of equipment and investments). Both layouts give the same income before taxes (52,000 €) and the same net income (36,400 €); only the EBIT subtotal changes: 31,800 € here and 42,300 € in the key. For an exam marked against this key, expect EBIT = 42,300 €.

Exercise 5

Exercise 5 — Income statement 2017 with discounts, payroll, depreciation, taxes and dividends

Original statement · T3_Financial Statements_EXERCISES.pdf, p. 2

Prepare the income statement of a company with the following information for 2017:

- *Gross sales during the year reached 2,002 million €. The company's trade policy gives a 3% discount to its Type A customers, those whose annual purchases from the company exceed 120 million €. Type A customers account for 40% of sales. Assume total sales = gross sales − discount.*
- *Purchases of raw materials amounted to 962 million €.*
- *The workforce consists of 7,800 employees. The average salary is 30,000 € per year, and the average spending on Social Security is 30% of the salary.*
- *Payments for other operating expenses amounted to 457 million €.*
- *In recent years the company has invested a total of 1,200 million € in fixed assets with an average lifespan of 10 years (linear depreciation, no residual value).*
- *The company obtained an international loan of 225 million € in 2015, at 6% interest along the whole year; the whole principal will be repaid in 2020.*
- *The corporate tax rate is 30%. The company has agreed to a dividend of 50%.*
- *Assume no change in stocks of raw materials or finished products during the year.*

Phase 1 — Setup: units and hypotheses

All amounts are expressed in **millions of euros (M€)**. Salaries are given per employee in €, so they must be converted: $1\ \text{M€} = 10^6\ €$.

Hypotheses:

1. "No change in stocks" means initial inventory = final inventory, so $\text{COGS} = \text{Purchases}$.
2. The 120 M€ threshold only defines who is a Type A customer; the statement already says these customers make 40% of the sales, so the discount applies to that 40%.
3. The loan principal is not an expense: only its interest enters the income statement [Slide 11].
4. The dividend of 50% is applied to the **net income** [Slide 23].

Phase 2 — Frames: line of each item

Item	Line
Gross sales − discount	Sales revenue (net)

Item	Line
Raw materials (no stock change)	Cost of goods sold
Salaries + Social Security	Operating expense
Other operating expenses	Operating expense
Depreciation of fixed assets	Operating expense [Slide 10]
Interest on the loan	Non-operating item [Slide 11]
Tax 30%	Income taxes
Dividend 50%	Distribution of net income (not an expense)

Phase 3 — Derivation

Step 1. Net sales

$$\text{Type A sales} = 0.40 \times 2{,}002 = 800.8 \text{ M€}, \qquad \text{Discount} = 0.03 \times 800.8 = 24.024 \text{ M€}$$

$$\text{Net sales} = 2{,}002 - 24.024 = 1{,}977.976 \text{ M€}$$

Step 2. Personnel expense

$$\text{Salaries} = 7{,}800 \times 30{,}000 = 234{,}000{,}000 \text{ €} = 234 \text{ M€}$$

$$\text{Social Security} = 0.30 \times 234 = 70.2 \text{ M€}, \qquad \text{Personnel} = 234 + 70.2 = 304.2 \text{ M€}$$

Step 3. Depreciation and interest

$$D = \frac{1{,}200 - 0}{10} = 120 \text{ M€/year}, \qquad I = 225 \times 0.06 = 13.5 \text{ M€}$$

Step 4. Income statement 2017

Line	Computation	M€
Net sales	Step 1	1,977.976
– Cost of goods sold (raw materials)		962.000
= Gross profit	$1{,}977.976 - 962$	**1,015.976**
– Personnel	Step 2	304.200
– Other operating expenses		457.000
– Depreciation	Step 3	120.000
= Income from operations	$1{,}015.976 - 881.2$	**134.776**
– Interest expense	Step 3	13.500
= Income before taxes	$134.776 - 13.5$	**121.276**
– Income taxes (30%)	0.30×121.276	36.383
= Net income	$121.276 - 36.3828$	**84.893**

Step 5. Distribution of the net income [Slide 23]

$$\text{Dividends} = 0.50 \times 84.8932 = 42.447 \text{ M€}, \qquad \text{Retained earnings} = 84.8932 - 42.4466 = 42.447 \text{ M€}$$

Pay-out 50%, self-financing 50%.

Phase 4 — Interpretation and checks

- **Applying the discount to all the sales** would deduct $0.03 \times 2{,}002 = 60.06$ M€ instead of 24.024: the 40% restriction matters.
- **The loan repayment in 2020** does not appear in 2017: repaying principal is a payment that is never an expense.
- **Exact values:** income taxes 36.3828 M€ and net income 84.8932 M€ (shown rounded to three decimals in the table).
- **Dividends** come after taxes and are not an expense: the income statement ends at the net income, and the 50/50 split is shown separately.

Check against the official key

OFFICIAL SOLUTION, T3_FINANCIAL STATEMENTS_SOLUTIONS.PDF, P. 6

Every line coincides with the key: discount 24.024; net sales 1,977.976; cost of goods sold 962; personnel 304.2; other operating expenses 457; depreciation 120; EBIT 134.776; interest 13.5; income before taxes 121.276; tax 36.3828; net income 84.8932; dividends 42.4466 and retained earnings 42.4466 (all in M€).

Exercise 6

Exercise 6 — Company X, first-year balance sheet and partly paid capital

Original statement · T3_Financial Statements_EXERCISES.pdf, p. 3

Company X was incorporated on 1/01/2017 with a subscription of shares worth 100 m.u., fully paid in cash. On 1/01/2017 it bought, with cash: a) an area (land) for 20 m.u.; b) equipment with a life of 10 years (straight-line depreciation) and a purchase price of 20 m.u.; c) raw materials for 20 m.u.

The company hires an employee for a total annual cost (salaries, Social Security, etc.) of 10 m.u. per year. This employee processes half of the raw material and sells it for a total of 50 m.u. Sales are the same every day of the year. The average collection period is 3 months.

The single tax rate on the profits of the company is 30% (paid on 31/12/2017). The company follows a strict self-financing strategy.

1. *Prepare the balance sheet at 31/12/2017, justifying and explaining the intermediate calculations.*
2. *Explain how the accounting statements would change if, instead of 100%, the shareholders had paid only 90% of the 100 m.u. subscribed.*

Phase 1 — Setup: data and hypotheses

Period: calendar year 2017. **Closing:** 31/12/2017.

Hypotheses:

1. The employee's cost is paid during the year.
2. Land has no useful life given and is not depreciated; the equipment has no residual value given (zero).
3. **Strict self-financing** [Slide 23]: the whole net income is retained (pay-out 0%).
4. The tax is paid on 31/12, so no tax liability remains.

Phase 2 — Frames: classification of every operation

Operation	Income statement 2017	Cash 2017	Open at 31/12/2017
Capital	–	+100	Share capital 100
Land	–	−20	Land 20
Equipment	Depreciation 2	−20	Equipment 20 − 2
Raw materials	COGS 10 (half consumed)	−20	Inventory 10
Employee	Personnel 10	−10	–
Sales	Revenue 50	+37.5	Receivables 12.5

Operation	Income statement 2017	Cash 2017	Open at 31/12/2017
Tax	8.4	-8.4	–

Phase 3 — Derivation (requirement 1)

Step 1. Depreciation and cost of goods sold

$$D = \frac{20 - 0}{10} = 2 \text{ m.u.}, \qquad \text{COGS} = \frac{20}{2} = 10 \text{ m.u.}, \qquad \text{Inventory} = 20 - 10 = 10 \text{ m.u.}$$

Step 2. Receivables from the collection period

Sales are spread evenly over the year and customers pay 3 months after buying. At 31/12 the sales of the last 3 months are still unpaid:

$$\text{Receivables} = 50 \times \frac{3}{12} = 12.5 \text{ m.u.}, \qquad \text{Collected} = 50 - 12.5 = 37.5 \text{ m.u.}$$

Step 3. Income statement 2017

Line	Computation	m.u.
Sales revenue		50.0
– Cost of goods sold	Step 1	10.0
= Gross profit		**40.0**
– Personnel		10.0
– Depreciation	Step 1	2.0
= Income from operations = income before taxes	no interest	**28.0**
– Income taxes (30%)	0.30×28	8.4
= Net income	$28 - 8.4$	**19.6**

All 19.6 is retained (strict self-financing).

Step 4. Cash at 31/12/2017

$$\text{Cash} = 100 - 20 - 20 - 20 + 37.5 - 10 - 8.4 = 59.1 \text{ m.u.}$$

Step 5. Balance sheet at 31/12/2017

Assets	m.u.	Liabilities and equity	m.u.
Land	20.0	Share capital	100.0
Equipment (20 − 2)	18.0	Retained earnings (net income 2017)	19.6
Non-current assets	**38.0**	**Equity**	**119.6**
Inventory (raw materials)	10.0	**Liabilities**	**0**
Accounts receivable	12.5		
Cash	59.1		
Current assets	**81.6**		
Total assets	**119.6**	**Total liabilities and equity**	**119.6**

Phase 3b — Derivation (requirement 2: only 90% paid)

Step 6. What changes

The shareholders **subscribed** 100 m.u. but **paid in** only $0.90 \times 100 = 90$ m.u. The remaining 10 m.u. is capital not yet paid (uncalled capital): the shareholders still owe it to the company.

- **Income statement: no change**. Revenues and expenses do not depend on how the capital was paid; there is no interest involved. Net income stays 19.6.
- **Cash:** 10 m.u. less: $59.1 - 10 = 49.1$ m.u. All the purchases (60) can still be paid in cash, since the company receives 90.
- **Equity:** the contributed capital is what the stockholders actually provided [Slide 13, Slide 17], so the subscribed capital is shown minus the unpaid part: $100 - 10 = 90$.

Assets	m.u.	Liabilities and equity	m.u.
Non-current assets	**38.0**	Subscribed capital	100.0
Inventory	10.0	– Uncalled capital (unpaid by shareholders)	(10.0)
Accounts receivable	12.5	Retained earnings	19.6
Cash	49.1	**Equity**	**109.6**
Current assets	**71.6**	**Liabilities**	**0**
Total assets	**109.6**	**Total liabilities and equity**	**109.6**

Both totals fall by exactly the 10 m.u. not paid.

Phase 4 — Interpretation and checks

- **Balance:** $38 + 81.6 = 119.6 = 100 + 19.6$ (requirement 1); $38 + 71.6 = 109.6 = 90 + 19.6$ (requirement 2).
- **Equity bridge:** equity = capital paid + net income retained: $100 + 19.6$ and $90 + 19.6$.
- **Profit vs cash:** after the purchases of 1/01, the cash generated during the year is $37.5 - 10 - 8.4 = 19.1$, while the profit is 19.6. Reconciliation: $19.6 + 2 + 10 - 12.5 = 19.1$, that is, profit + depreciation (expense, never a payment) + materials consumed (expense whose payment was made on 1/01 with the purchase) − sales not collected yet.
- **Presentation note:** some textbooks show the 10 m.u. owed by shareholders as an asset instead of deducting it from capital; then both totals stay at 119.6. The deduction used here is the one consistent with "contributed capital" in [Slide 17]. Both readings give the same net equity provided (109.6) and leave the income statement unchanged.

Check against the official key

OFFICIAL SOLUTION, T3_FINANCIAL STATEMENTS_SOLUTIONS.PDF, PP. 7-8

Requirement 1 coincides with the key: sales 50; cost of goods sold 10; personnel 10; depreciation 2; EBIT 28; tax 8.4; net income 19.6; cash 59.1; receivables 12.5; inventory 10; land 20; equipment 18; total assets 119.6. The key does not solve requirement 2 (90 % paid in), so that part of the solution is not covered by it.